

Received 3 October 2024, accepted 26 November 2024, date of publication 19 December 2024,
date of current version 30 December 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3510456

SURVEY

Artificial Intelligence Driven Smart Farming for Accurate Detection of Potato Diseases: A Systematic Review

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This work was supported by the Natural Sciences and Engineering Council of Canada.

ABSTRACT Agriculture can ensure food security and enhance monetary benefits if practiced with modern technologies and supported with artificial intelligence (AI). Modern advancements in farming practices have revolutionized the production of food vegetation. However, crop cultivation faces several threats including insect and pest attacks and disease infections on plant leaves. For example, one of the most consumed foods vegetables universally—potatoes, are vulnerable to diseases like Late Blight (LB), Early Blight (EB), and others. These infections must be controlled to enhance food quality and yield. Conventional disease detection techniques are slow and depend on human involvement, which may be laborious and erroneous. However, AI tools, for instance, Machine Learning (ML) and Deep Learning (DL), offer precise and well-timed solutions for disease detection, classification, and eradication. A comprehensive review of literature has been conducted by examining over 400 articles to focus on 72 studies including 14 reviews publications on ML and DL models about potato disease forecasting using different techniques. It highlights the need for proficient disease control by integrating image and climate data. It further aids in addressing challenges like data availability and geographical variations. It has been learned that image-processing techniques overwhelm the existing research and have the potential to integrate meteorological data. The most widely used algorithms incorporate Support Vector Machine (SVM), Random Forest (RF), Convolutional Neural Network (CNN), and MobileNet with accuracy rates between 64.3 and 100%. The importance of accurate disease detection and eradication has been reported for food security, financial stability, and sustainable farming practices. Progressions in disease forecasts aid farmers in making informed decisions, minimizing crop losses, and reducing pesticide use through targeted application of agrochemicals with the use of AI-driven variable rate sprayers. This leads to healthier crops, market stability, and a more sustainable farming environment.

INDEX TERMS Artificial intelligence, deep learning, food security, machine learning, potato disease forecasting.

NOMENCLATURE

AI Artificial Intelligence.

ANN Artificial Neural Network.

BPNN Back Propagation Neural Network.

CNN Convolutional Neural Network.

DL Deep Learning.

DT Decision Tree.

EB Early Blight.

ELM Extreme Learning Machine.

The associate editor coordinating the review of this manuscript and approving it for publication was Yin Zhang¹.

IoT	Internet of Things.
KNN	K-Nearest Neighbors.
LB	Late Blight.
LDA	Linear Discriminant Analysis.
LR	Logistic Regression.
LSTM	Long Short-Term Memory.
ML	Machine Learning.
MLP	Multi-Layer Perceptron.
MPE	Mean Percentage Error.
MSE	Mean Square Error.
M-SVM	Multi-Support Vector Machine.
NB	Naive Bayes.
NN	Neural Network.
PA	Precision Agriculture.
RF	Random Forests.
RM	Regression Models.
RMSE	Root Mean Square Error.
RNN	Recurrent Neural Network.
RS	Remote Sensing.
SVM	Support Vector Machine.
SVR	Support Vector Regressor.
UAV	Unmanned Aerial Vehicle.

I. INTRODUCTION

Agriculture contributes to alleviating poverty and fosters economic growth of individual farmers and countries' gross domestic product (GDP). It is crucial for ensuring food security and sustaining the livelihoods of the planet Earth [1]. Food insecurity poses a significant risk, leading to numerous challenges such as malnutrition and economic instability in the masses. Therefore, on a global scale, more food production is the most powerful weapon to deal with food insecurity [2]. In rural areas, agriculture is a primary livelihood, where about 80% of people are directly and indirectly involved in agricultural activities [3].

Among the different food crops cultivated globally, potatoes stand as the most favored one. Regarding human consumption, potatoes are the third most substantial food crop after cereals (rice and wheat) worldwide [4]. Potatoes form a staple food for millions and are rich in essential vitamins, minerals, nutrients, proteins, and energy. A report published by Statista says that in 2021, over 376 million metric tons of potatoes were produced globally [5]. Economically, potato crops essentially contribute to the agricultural GDP of many countries. For instance, the U.S. potato industry alone generated an estimated \$100.9 billion in economic activity in 2021, supporting over 714,000 employees and contributing \$34 billion in wages [6].

Aligning with the Sustainable Development Goals (SDGs), potatoes contribute to several food security targets. They are essential for promoting SDG 2: Zero Hunger [7], by providing a reliable and nutritious food source that can help reduce starvation and undernutrition. Moreover, sustainable potato cultivating practices can support SDG 12: Responsible Consumption and Production [8], by advancing efficient

utilization of natural resources and reducing food waste. In general, the role of potatoes in enhancing food security and supporting economical agricultural practices underscores its worldwide importance.

A. CHALLENGES FACING POTATO CULTIVATION

Various potato species with different sizes, colors, and shapes have different benefits and can be developed in a variety of climates and are a crucial food source for many nations. However, global potato production faces significant challenges from several diseases and disorders during their cultivation period. Around 20-40% of overall food crops are lost due to diseases and pest attacks globally [9] as such, worldwide, 32% of potatoes are lost annually [10].

Numerous factors are responsible for the spread of infections in crops. The disease triangle model [11] is taken into consideration as the principal behind the disease development. As for this model, three predominant components including a plant, favorable environmental conditions, and a pathogen cause disease. An infection evolves when these three factors co-exist, which further causes damage to the plants and reduces crop yields as demonstrated in FIGURE 1. The plants remain healthy in the absence of any of these factors [12].

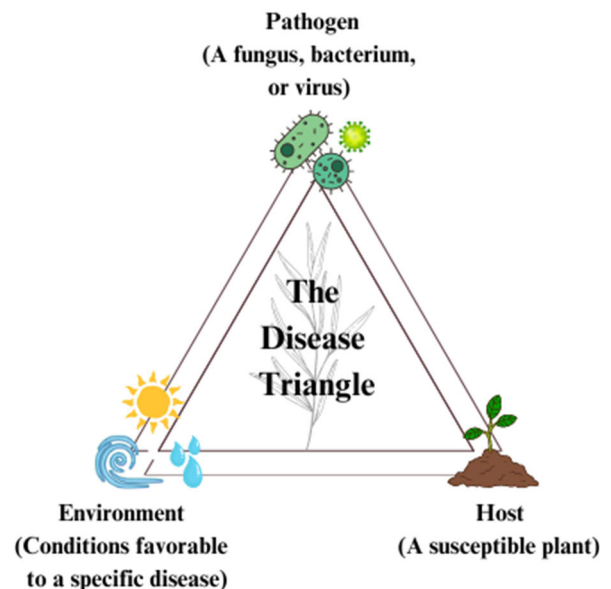


FIGURE 1. The disease triangle: interplay of pathogen, environment, and host in plant diseases.

Potato plants are vulnerable to numerous diseases [13], which can noticeably reduce crop yield and quality [14]. The diseases are classified into two classes: biotic and abiotic diseases as displayed in FIGURE 2 (top).

Biotic diseases, also known as infectious diseases, are caused by living organisms such as fungi, bacteria, viruses, tubers, insects, and mites that externally attack plant shoots. On the other hand, abiotic diseases are caused by non-living

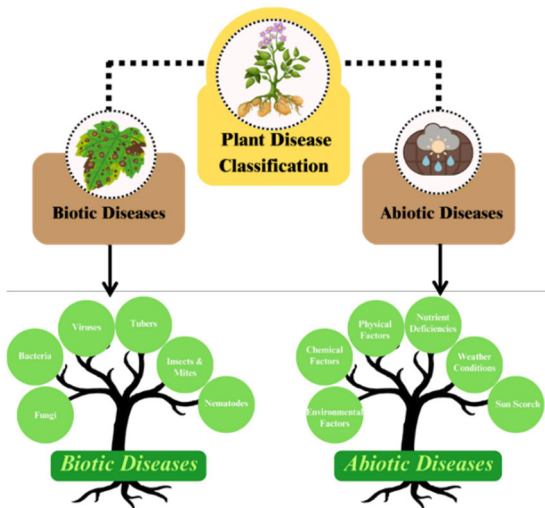


FIGURE 2. Potato plant disease classification: biotic and abiotic diseases (top) and overview of potato plant diseases (bottom).

organisms including environmental factors, chemical factors, physical factors, nutrient deficiencies (nitrogen, phosphorus, and potassium), and weather conditions are shown in FIGURE 2 (bottom) and are also known as non-infectious diseases.

B. COMMON POTATO DISEASES, THEIR CLASSIFICATION, AND SYMPTOMS

A comprehensive Sankey diagram as presented in FIGURE 3 highlights the extensive range of potato diseases detected by various studies from 2007 to 2024. This diagram illustrates the intricate connections between the general category of plant diseases, the specific crop of potatoes, and the detailed list of diseases that have been the focus of research over the years. The two thick lines driving from “Plant Disease” to “Crop Potato” and after that to “Early Blight” and “Late Blight” (EB and LB, respectively) show that these two diseases are the most common among the ones considered. These thick lines imply that a considerable number of analysts universally have concentrated their endeavors on foreseeing and overseeing these diseases. The variety of thinner lines connecting “Crop Potato” to other diseases such as Bacterial Wilt, Common Scab, Black Scurf, and others reflect the differences of pathogens influencing potatoes and the broader scope of investigation aimed at understanding and mitigating these threats. Each line represents the linkage between the plant diseases affecting potatoes and the specific type of disease, emphasizing the comprehensive nature of the studies conducted over a long time. This chart not only underscores the significance of EB and LB within the context of potato disease research but also gives a visual outline of the broader range of diseases that have been inspected, subsequently offering a holistic view of the endeavors made to ensure the health and productivity of potato crops globally.

As reported in the literature, the major potato diseases include LB, caused by water mold infestans [15], and EB,

avored by warm and wet weather conditions [16]. The agent responsible for LB is a fungal phytopathogen, known as *Phytophthora Infestans* and the *Alternaria Solani*/*Alternaria* is responsible for EB [17]. In summary, potato LB is caused by humid, warm weather along with frequent precipitation and moderate temperature (15°C - 20°C) [18]. EB disease is favored by drier weather conditions with dry and warm weather, temperature (21°C - 29°C), high humidity, and lack of rainfall. These diseases are notorious for their impact on potato crop cultivation worldwide.

To get rid of diseases, pesticides are overused which further exploits human health and the environment [19]. Throughout human history, plant disease occurrences have resulted in widespread human disease, poverty, and famine [20]. The disease is a major cause of crop yield losses around the world and a considerable reason for food insecurity in developing countries and health losses in developed countries [21]. The insets in FIGURE 4 illustrate a visual of the most common potato diseases including EB, LB, and healthy, highlighting the stark differences and the impact of diseases on plant health.

Literature has focused on EB, LB, and other diseases. The pie chart in FIGURE 4 illustrates the focus of research publications on potato disease predictions from 2007 to 2024 [15], [16], [17], [18], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [112] by categorizing them into four groups. The largest segment (44.4%) represents [15], [16], [17], [18], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [86], [87], [88], [112] addressing both LB and EB and highlighting the significant impact of these diseases. This is followed by research on other diseases comprising 18.1% [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58] and indicating a broad interest in diverse potato disease prediction. Publications exclusively focused on LB account for 29.2% [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79] underscoring its critical importance. The smallest category represents 8.33% [80], [81], [82], [83], [84], [85] which is dedicated solely to EB. This distribution underscores the concentrated research efforts on major blight diseases while also acknowledging other impactful potato diseases. This visual representation emphasizes the importance of identifying and addressing these diseases to maintain crop health and productivity.

C. POTATO DISEASE DETECTION APPROACHES

Approaches to potato disease detection are divided into traditional and advanced nature. Traditional approaches have been replaced and combined with advanced technologies due to

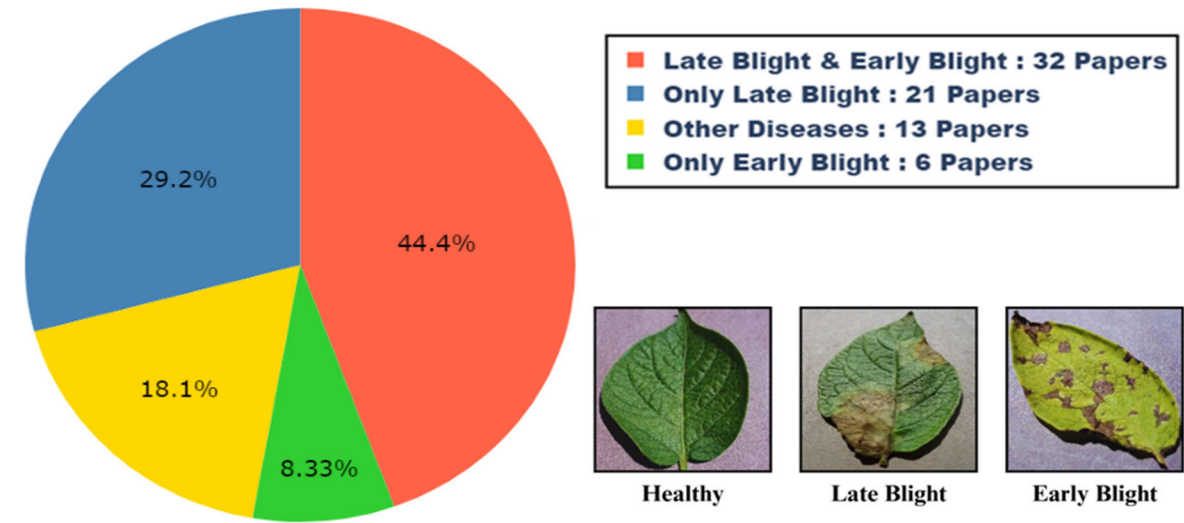


FIGURE 3. Distribution of research works (2007-2024) on prediction of potato diseases shown in the inset images.

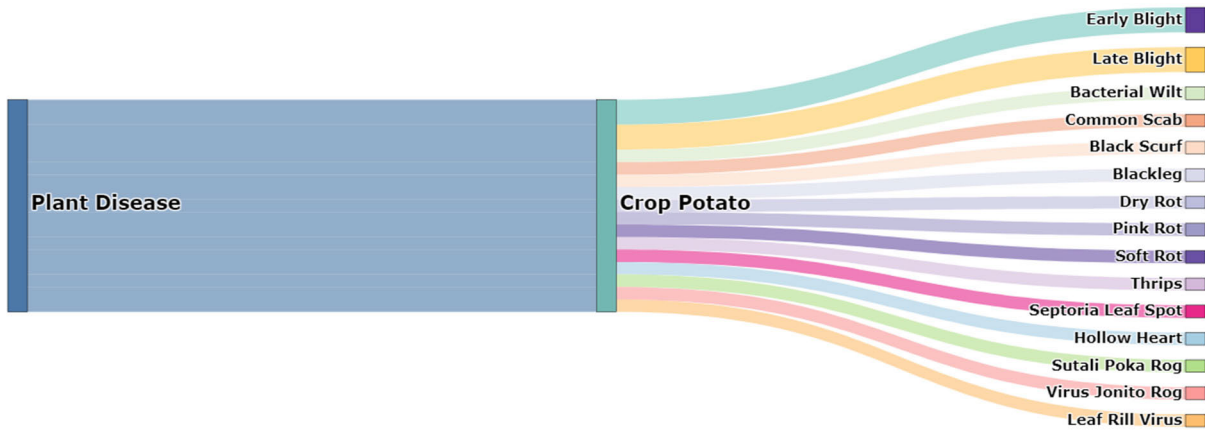


FIGURE 4. Sankey diagram illustrating the range of potato diseases identified by various studies from 2007 to 2024 [15], [16], [17], [18], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [112].

automation, error-free disease detection, reduced labor work, and more effectiveness.

1) TRADITIONAL METHODS

Before the advent of the technological era, researchers, farmers, and agricultural experts relied on traditional approaches to detect potato diseases in crops. This involved utilizing specialized information from various specialists within the field, visually analyzing plant foliage, and conducting manual assessments. However, the execution of these ordinary strategies displayed a few challenges. Firstly, hiring personnel for field surveys is expensive, and enough experts in infection identification do not exist. Moreover, conventional strategies are labor-intensive, time-consuming, and impractical for surveying large areas of farmland in totally different areas.

2) ADVANCED METHODS

With progressions in technology, more advanced methods have been developed to speed up and improve crop disease identification. These novel strategies utilize information from distinct sources, like IoT (Internet of Things), UAV (Unmanned Aerial Vehicle), RS (Remote Sensing), drone imagery, field surveys, weather stations, and the web which allows for real-time data collection and monitoring in rural settings. ML and AI center on the creation of predictive models based on historical information to improve and progress the exactness of disease detection. The incorporation of automated frameworks marks a critical step in streamlining processes, diminishing manual efforts, and accelerating disease identification. These advanced methodologies have changed infection detection, making the method quicker, more efficient, and cost-effective. The transition from

conventional to advanced technologies is persuaded by the need for early forecast of distinct crop diseases.

D. KNOWLEDGE GAP, NEED, AND OBJECTIVES OF THIS REVIEW PUBLICATION

Potatoes are a fundamental food crop and source of food for a wide range of populations [89] and are crucial for food security [90]. Potato disease modeling has improved with advanced technologies, but there are still important research gaps, especially in using time series forecasting. Traditional methods of disease detection in potatoes depend on manual checks, which is hard work and needs expert knowledge, causing delays in finding and responding to diseases. Most research focuses on identifying diseases using images [17], [18], [19], [20], [21], [23], [25], [47], [48] helpful but has its limits in early detection. The lack of complete data, especially weather data, makes the problem worse, affecting crop health and food security, and potentially harming people who eat diseased potatoes. This research gap makes it hard to develop efficient and accurate methods to forecast diseases.

The upsides of advancing disease detection in potatoes extend beyond enhancing crop yields. Potatoes benefit in the prevention and management of several human diseases, including, diabetes, dyspepsia, cardiovascular, and cholesterol diseases [20]. Healthy potato crops can thus have a direct impact on human health, providing a nutritious food source that supports overall well-being. Moreover, improving potato production can create jobs, improve farmer livelihoods, contribute to national prosperity, and boost the economy.

According to Masanobu Fukuoka [91] “The ultimate goal of farming is not the growing of crops, but the cultivation and perfection of human beings”. Over time, various practices for disease classification have already been implemented [92]. Traditional practices can be tedious and costly. Thus, there is a decisive necessity to automate disease management more effectively and economically using ML and Artificial Intelligence (AI) which ensure timely and accurate disease detection and facilitate farmers to manage their crops efficiently.

Modern technology [93] poses reliable and flexible advancements for disease detection by introducing new methods to safeguard potato crops. ML [94] has appeared as a remarkable tool that gives convincing results in identifying diseases by analyzing extensive data precisely and promptly that may be overlooked by human inspectors. However, the application of these technologies in agriculture is still new and there are many challenges to overcome to fully recognize their abilities.

Researchers employed several ML and DL models including (SVM) [21], [30], [35], [36], [39], [44], [45], [49], [52], [55], [58], [66], [67], [73], Naïve Bayes (NB) [25], [35], K-Nearest Neighbor (KNN) [25], [30], [35], [36], [78], Decision Trees (DT) [35], [78] Random Forest (RF) [21], [29], [54], [63], [64], [80], [83] Logistic Regression (LR) [21], [30], [35], [36], [70], Artificial Neural Network (ANN) [30], [57], [65], [74] and Convolutional Neural Network (CNN)

[18], [19], [20], [24], [26], [27], [29], [31], [32], [36], [38], [41], [42], [43], [48], [49], [50], [51], [52], [53], [64], [77], [78], [80], [95], for the classification of potato diseases and providing solutions to mitigate them. With this much information available in the literature, there is a need to understand the importance of potatoes for global food security, conduct a thorough review of existing methods for potato disease management, and guide policymakers and researchers for improvements in efficient potato cultivation to ensure its role in mitigating global food security issues.

This review paper provides information on how crop diseases threaten potato crops and directly impact food security by causing massive yield losses. Manual inspection has its limitations. Therefore, there's a need for better, automated methods to predict diseases, helping to reduce crop losses and ensure food security. After reviewing existing research, we found a major gap in not having enough studies utilizing weather data. Most research relies on images, but we need to explore time series data, including weather and plant conditions. This gap is a chance for more research. This paper aims to close the gap between manual and advanced techniques, showing new methods for early disease detection and management. These advanced techniques could lead to better results, helping future studies, improving resource use, and enhancing food security [96].

In recent years, potato cultivation has been drastically influenced by diseases [97]. Early and timely detection of outbreaks can prevent significant financial losses for farmers. Scientists have suggested that forecasting algorithms provide early predictions to help in preventing and controlling various diseases. The objective of this study is, i) to understand the importance of potatoes for global food security, ii) provide a thorough review of existing methods used by researchers globally for potato disease management by focusing on advancements in modern technology, and iii) suggest improvements in efficient potato cultivation to ensure its role in mitigating global food security issues.

II. METHODOLOGY

Literature on potato cultivation regarding its diseases and their detection, published during the last 18 years (i.e., 2007-2024) was reviewed for this article FIGURE 5. A pie chart, shown as an inset in FIGURE 5 shows the distribution of distinct datasets used in the reviewed studies over the period from 2007 to 2024. The chart classifies the datasets into three categories: imagery data (51 publications), climate data (18 publications), and other data sources (3 publications). Out of these, 70.8% of the research in the potato disease identification field utilized imagery data, 25% used climate data, and 4.2% relied on other types of data. In total, 72 publications were reviewed and plotted for their annual distributions; for example, 15 publications were from the year 2023. Further, this paper also includes 14 review publications focused on potato disease prediction. These research and review publications have global representation, prioritizing identification and prediction of potato crop diseases using

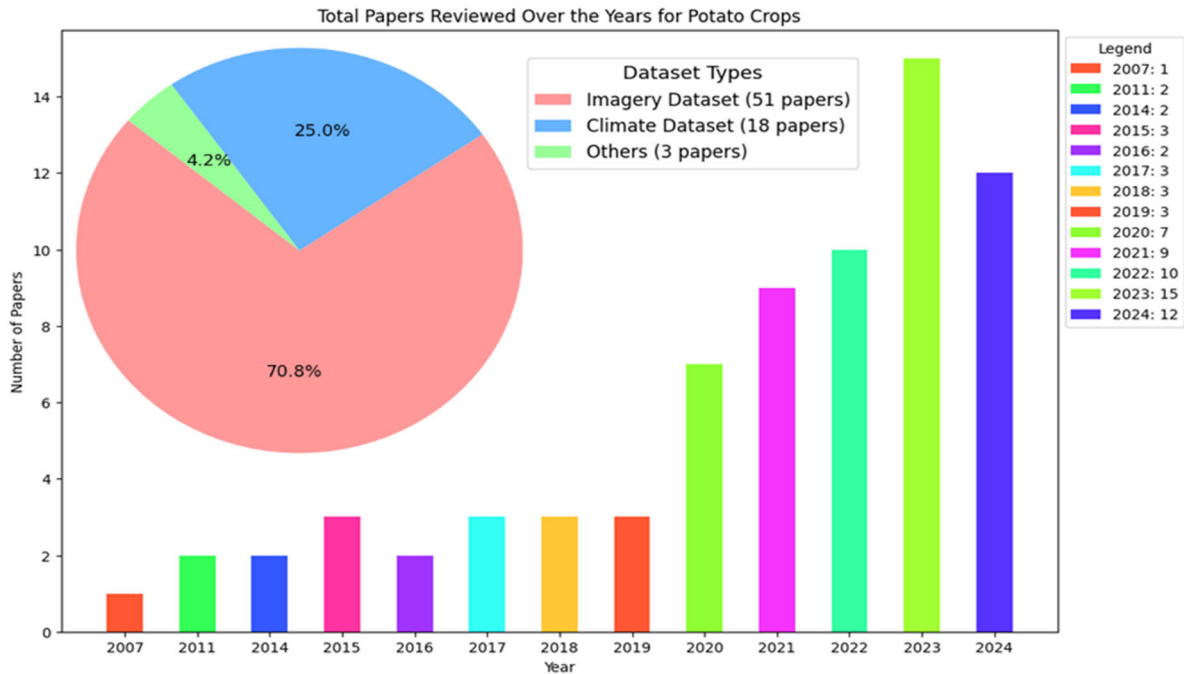


FIGURE 5. Pie Chart (Left): Distribution of dataset types used in reviewed studies (2007-2024): Imagery (51 publications), Climate (18 publications), and Others (3 publications). Bar Graph (Right): Total number of publications reviewed on potato crops from 2007 to 2024.

meteorological data and imagery data through advanced technologies including ML, DL, AI, RS, and other advanced methodologies. Most of the image data in existing research has been acquired from web scrapping tools, Kaggle, open-source repositories, plant village dataset, plant leaf disease, the internet, and manually captured photographs in fields using drones, and hyperspectral cameras.

Different keyword searches have been performed for conference and research publications using several search engines including Scopus, IEEE Access, Springer, Google Scholar, Connected Papers, Elicit, Science Direct, MDPI, and others. The search keywords included potato, crop disease, late blight, early blight, machine learning, deep learning, artificial intelligence, potato disease detection, leaf disease detection, sustainability, and some other keywords were used in this review. Initially, the total collections included more than 400 publications. The publications then underwent thorough examination, sorted according to their similarity and investigation before 72 publications were included in this investigation.

III. MAJOR FINDINGS

Technological advancements in plant disease detection include capturing good quality images, image processing using computer software, categorizing leaf infection symptoms with ML algorithms for automating disease detection from the symptoms on new leaves, and benefiting from AI for smart farming. Advanced computing in agriculture, using ML, AI, and DL, enhances precision agriculture (PA) by analyzing data from weather stations, sensors, and satellite imagery.

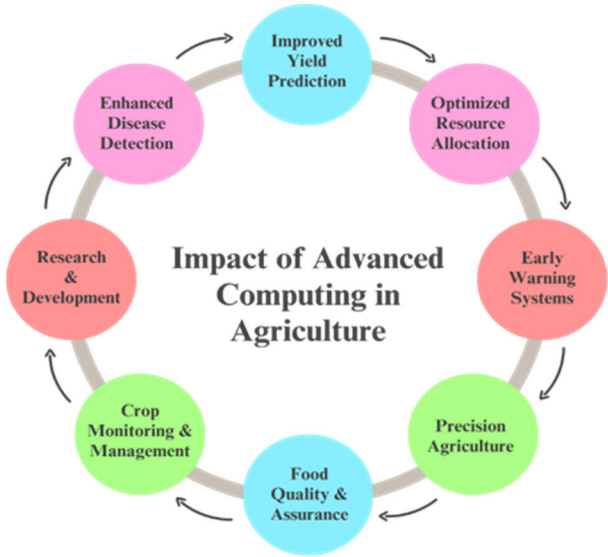


FIGURE 6. Impact of advanced computing (ML, DL, and AI) in agriculture.

These technologies optimize crop yields, predict diseases, optimal resource management, disease early warning, PA, enhanced food quality, crop monitoring, more research endeavors in agriculture, and automate irrigation, leading to increased efficiency, reduced costs, and sustainable farming practices FIGURE 6.

A. DATA ACQUISITION AND PROCESSING

The process of potato crop disease classification (FIGURE 7) begins with the collection of diverse input data, including

images of potato crop leaves, which may be taken from fields, existing databases, hyperspectral cameras, UAV, online sources, remote sensing, and plant village database or time series data such as temperature, humidity, and other environmental factors obtained from weather stations. These inputs undergo pre-processing to enhance quality and remove noise, with image pre-processing including resizing, normalization, augmentation [15], [29], [47] and filtering, whereas time series data may be smoothed and normalized.

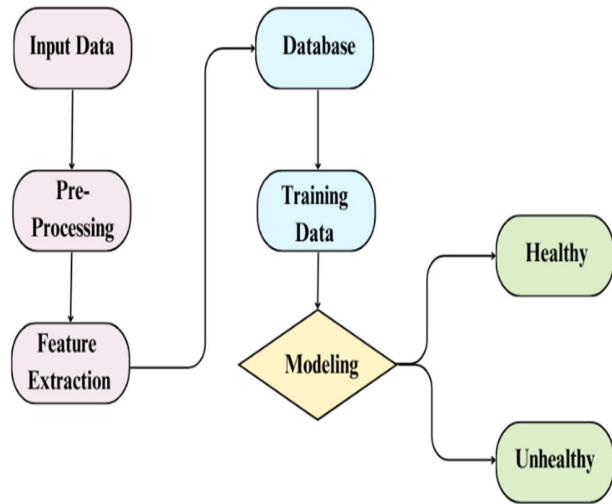


FIGURE 7. Flow chart of potato crop disease classification process using multimodal data.

Following pre-processing, feature extraction is performed on both data types: image features such as color, texture, and shape are analyzed, whereas time series highlights might incorporate statistical measures, trends, and patterns over time. The extracted features are then organized into a structured format, such as feature matrices for images and time series data and stored in a database for proficient management. This organized data is used to train ML [21], [25], [35], [40], [45], [54], [55], [60], [65], [66], [73], [75], [77], [78] and various DL models [15], [16], [17], [18], [19], [20], [21], [26], [27], [28], [33], [37], [41], [46], [47], [49], [51], [55], [62], [66] such as SVM, CNN, DT, NB, and KNN. CNN for image data and Recurrent Neural Networks (RNN) or other time series models for temporal data. These models learn to detect patterns and links between the features and disease categories. Once trained, the models then classify new and unseen data into healthy or unhealthy categories based on the learned features. The results indicate whether the potato crops are healthy or unhealthy and are then used for further analysis, reporting, or decision-making in crop management.

B. DEEP LEARNING, MACHINE VISION, AND ARTIFICIAL INTELLIGENCE FOR DISEASE SYMPTOM DETECTION

This section highlights the importance of ML, DL, and AI techniques in disease detection. This further includes the

processes involved, datasets and data sources employed, and distinct approaches implemented for disease detection.

1) MACHINE LEARNING

Machine learning is a branch of AI that empowers computers to memorize data and improve their performance over time without being explicitly programmed. It involves feeding enormous amounts of information into models that can recognize patterns and make decisions. There are three main types of ML consisting of supervised learning, unsupervised learning, and reinforcement learning as demonstrated in FIGURE 8.

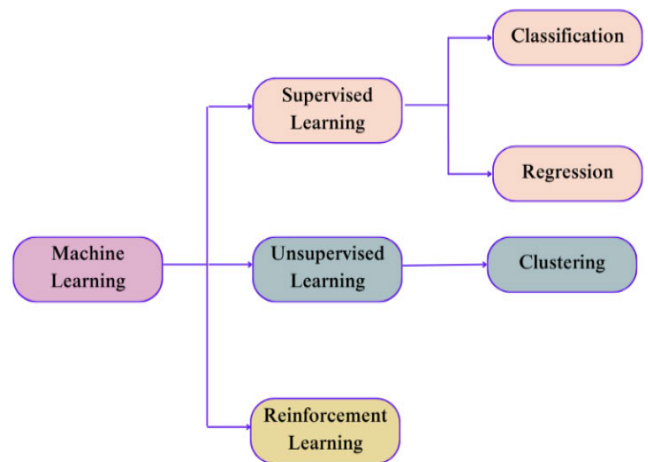


FIGURE 8. Overview of machine learning types: supervised, unsupervised, and reinforcement learning.

In supervised learning, the model is trained on labeled data and used in tasks like classification and regression. Classification algorithms are used when the output variable is categorical. These include linear classifiers, SVM, DT, NB, KNN, and RF. Regression models are useful for forecasting numerical values based on various data points. Linear regression, LR, and polynomial regression are some popular regression algorithms. On the other hand, unsupervised learning deals with unlabeled data and intends to find hidden within the data, commonly used in clustering.

Clustering is a data mining technique for grouping unlabelled data based on similarities or differences. In addition, reinforcement learning includes training models through a framework of rewards and punishments. This sort is often applied in scenarios requiring decision-making. ML strategies are the foundation of advanced computing by affecting a wide range of areas from healthcare to farming.

2) ARTIFICIAL INTELLIGENCE AND DEEP LEARNING

AI includes creating frameworks that can perform tasks requiring human intelligence, such as language detection, pattern recognition, solving problems, and decision-making. AI incorporates various subfields, including ML and DL. DL, which is a further subset of machine learning, deals with neural networks having an 'n' number of layers to

analyze large amounts of data. DL is used in fields such as image and speech recognition, natural language processing, and autonomous driving. DL models are inspired by the structure and function of the human brain. These models can automatically extract features from raw data and enhance the capability to manage unstructured data.

C. LITERATURE REVIEW

Recent review publications on potato disease detection have extensively covered various technological approaches. A review of image processing techniques [98], [99] outlines methods like thresholding, edge detection, and segmentation, emphasizing their application in identifying diseases such as EB and LB while discussing challenges in image acquisition and pre-processing. Deep learning approaches, detailing the use of CNNs and ML [100], [101], [102] highlight high accuracy and potential for real-time detection, though noting the need for substantial computational resources and large annotated datasets. Additionally, a review of ML techniques [103] delves into methods such as SVM, RF, and KNN, comparing their performance and efficacy in disease diagnosis. Together, these reviews [104], [105], [106], [107], [108], [109], [110], [111] underscore the advancements and ongoing challenges in utilizing image processing, deep learning, and machine learning for effective potato disease detection.

1) IMAGE PROCESSING-BASED POTATO DISEASE DETECTION

According to Nishad et al. [22] in 2022, the authors proposed a novel approach using VGG16, VGG19, and ResNet50 to identify three classes of potato diseases including EB, LB, and healthy. The analysis had been proposed on imagery data (2580 images) acquired from 2 distinct sources (D1: Plant village and D2: Mendeley dataset). K-means clustering and data augmentation are performed. VGG16 outperforms 97% accuracy, followed by VGG19 and then ResNet with 95% and 67% accuracy, respectively. In 2023, Kasani et al. [23] conducted a study to classify potato diseases into two classes (standard or unhealthy). The imagery data composed of unhealthy (EB and LB), and standard (healthy) leaves were taken from a repository available on Kaggle for examination using the CNN model with 96.8% accuracy. Pasalkar et al. [17], Sai Ponnuru and Amasala [24], and N. Sagar et al. [25] from 2024, used a CNN-based approach for the image classification of infected and healthy leaves for EB, LB, and healthy potato leaves. The dataset from Kaggle called the new “plant disease dataset” (PDD) was taken with 600 images and resulted in 97.4% accuracy for the study [17], whereas the Plant Village dataset from Kaggle was taken for the study [24], [25] and achieved an impressive accuracy of 99.54% and 97.82%, respectively. In the same year, Khan et al. [26] also proposed a CNN-based model for the identification of potato diseases (EB, LB, and healthy) using imagery data. The dataset of 4500 images was sourced from three sources D1: 2152 images, D2:

1500 images, and D3: 848 images (D1: plant village dataset, D2: another repository, and D3: PLD) and obtained an overall average accuracy of 97.33%. Other metrics like recall, precision, and F1-Score had also been observed. A distinct image classification by Catal Reis and Turk [27], carried out in 2024, proposed a novel DL model called MDSCIRNet (Multi-head Attention Mechanism Depthwise Separable). The MDSCIRNet was further integrated with SVM, LR, AdaBoost, and RF algorithms for hybrid methods. The imagery dataset of 1500 images for three classes of potato diseases such as EB, LB, and healthy was taken for classification from “Potato leaf disease datasets” and a maximum accuracy of 99.33% was achieved for MDSCIRNet with SVM classifier. According to Rahman et al. 2023 [51], 15 classes of different diseases (belonging to Potato, tomato, and bell pepper) were predicted using the Plant Village dataset consisting of 54303 images. For classification, 10 distinct models were used, including DenseNet201, DenseNet121, NasNetLarge, Xception, ResNet152v2, EfficientNetB5, EfficientNetB7, VGG19, MobileNetV2, and hybrid model (EfficientNetB7 and ResNet152V2). The highest accuracy of 98.67% was obtained by DenseNet201 with a loss of 0.04. Anim-Ayeko et al. 2023 [28], focused on predicting EB, LB, and healthy classes for both potato and tomato crops using imagery data having 6652 images (2152: potato and 4500: tomato). The ResNet9 had been implemented with an accuracy of 99.25%. Other performance metrics like rec, pre, and F1-Score have also been observed. Pineda Medina et al. 2024 [29], developed an offline mobile application for detecting potato diseases into three different classes (EB, LB, and healthy). A DNN approach named MobileNetV2 had been utilized for 1000 images and an accuracy rate of 98.7% was achieved. Shrivastava et al. [30] for the detection of three classes of potato diseases using image classification was carried out in 2023. The VGG19 and ODCNN models were adopted for the analysis. Two distinct datasets (D1: Plant village with 2162 images and D2: PDLN with 4071 images) were collected from an open-source dataset repository on Kaggle. For D1 and D2, accuracy rates of 98.26% and 99.22%, respectively were obtained. For EB disease identification at different growth stages, Afzaal et al. [16] conducted a study in 2021 using 5199 images of healthy and disease categories from four fields. Three different CNN models: GoogleNet, VGGNet, and EfficientNet were considered for classification. The achievable maximum F1 score was 0.84 – 0.98.

An insightful strategy was proposed by Shabrina et al. 2024 [52], for a new dataset composed of 3076 images for the classification of potato diseases into seven distinct categories, including leaves attacks by viruses (532 images), bacteria (569 images), fungi (748 images), pests (611 images), nematodes (68 images), phytophthora (347 images), and healthy leaves (201 images) in Indonesia. The dataset was made accessible in a Mendeley data repository called “Publicly and freely available”. In addition to this, three DL models: EfficientNetV2B3, MobileNetV3-Large, and VGG16, were

selected and analysis were performed with and without data augmentation. With data augmentation, the results were similar. After analysis, with the composed dataset, a maximum of 73.63% accuracy was achieved with the EfficientNetV2B3 model. In future to make the dataset more reliable and diverse, pictures are recommended to be taken outside of Indonesia. Sharma et al. [48] suggested K-means clustering and ML models in 2021 for the image classification of three diseases, such as EB, LB, and healthy, in potato crops. The study implemented four different ML models such as SVM, KNN, NB, and DT and achieved a remarkable accuracy of 92.9% for the SVM model. In 2024, Rathore et al. [31] focused on the prediction of EB, LB, and healthy classes based on 5000 labeled pictures of potato leaves. The dataset was taken from an open-source "Plantify Dr Dataset" database. Three ML algorithms, RF, KNN, and NB were employed and obtained an accuracy of 99%, 85%, and 80%, respectively. Luong 2024 [53], proposed a novel custom model based on CNN known as the ConvNeXtSmall model. The imagery dataset was gathered from numerous websites and validated by BARI (Bangladesh Agricultural Research Institute) consisting of 451 images in total. Six potato diseases (Common scab, black scurf, blackleg, dry rot, pink rot, and miscellaneous) and healthy leaves were examined with the proposed model and achieved 99.49% accuracy. The proposed model was also then compared with several CNN models. For plant disease identification, Jha et al. 2024, [32] proposed DNN-based ensemble models which integrate ResidualNet, MobileNet, and Inception models. The disease classification was carried out for infected (EB and LB) and healthy classes based on the Plant Village dataset having 857 images. The ensemble model resulted in 98.86% accuracy which should be higher than individual DNN models. In 2021, Kumar Shukla et al. [54] introduced a DFS (Disease Forecasting System) for the identification of diseases in potato crops. The system was developed by the integration of K-means clustering, CNN, and SVM model and achieved 97.90% accuracy. Ghosh et al. 2023 [55], utilized healthy and afflicted specimens of potato leaves for disease diagnosis. Three CNN-based models: VGG19, DenseNet121, and ResNet50 were employed for image analysis and observed a maximum accuracy of 98.77% with a loss function of 0.0280 for VGG19, followed by ResNet50 and DenseNet121.

Arshaghi et al. 2023 [56] proposed a convolutional neural network model to predict five different potato diseases, including healthy plants. Using over 5000 images, the model achieved an accuracy of 98.65%. The researchers compared the proposed model with other established CNN models like AlexNet, GoogleNet, and VGG, highlighting the superior performance of the proposed model. Kang et al. 2023 [50] used a lightweight NN model to classify potato diseases. The imagery dataset was used consisting of 5450 images. The model achieved an accuracy of 93% for EB and LB predictions. In 2023, Samatha et al. [57] IoT with advanced image processing to distinguish healthy and unhealthy potatoes. An M-SVM model with CNN and DNN models was

implemented and an accuracy of over 99% was observed. In 2023, Bonik et al. [33] used a DL approach to classify potato plants as healthy or affected by EB and LB diseases. A dataset of 3561 images was fed to the CNN model and achieved an accuracy of 94.2%. In 2023, Sharma et al. [34] focused on predicting both EB and LB in potatoes. The study utilized a CNN model with three different activation functions (Elu, Swish, and ReLU) on a dataset of 1722 images, attaining a high accuracy of 98% with a swish activation function loss of 0.04. In 2022, Shi et al. [64] used drones equipped with hyperspectral cameras i.e., UAVs to capture images of potato fields in China. The imagery data was then applied to predict LB. By deploying the CropDocNet DL model, a high accuracy of 98.09% was acquired. Verma et al. 2023 [15] used data augmentation on a dataset of 1500 images. A CNN model was used to predict EB and LB in potato crops with an accuracy of 97%. Garcia-Ariza et al. 2020 [65] used ML models to predict LB in potato crops. The dataset of 320 images was deployed to K-means and hierarchical clustering algorithms. The model accomplished an average accuracy of 68.24%. In 2014, Dutta et al. [83] embraced RS indices such as NDVI and LSWI to foresee LB in potatoes. The dataset was composed of satellite images. In 2017, Patil et al. [49] used ML and DL-based image processing to predict EB and LB in potatoes. A dataset of over 892 images was employed for SVM, RF, and ANN models. The maximum accuracy of 92% was achieved using the ANN model. In 2011, Shankar Ray et al. [66], used hyperspectral data to predict LB in potatoes by analyzing vegetation indices from Nijjarpura Village, India. In 2022, Kumar and Shukla [35] used ML models like SVM, KNN, ANN, and LR on a dataset of 1931 images (D1: Plant village dataset and D2: Other images). A maximum accuracy of 99.75% with the SVM model was observed. Rayhan Asif et al. 2020 [46] compared CNN models such as AlexNet, ResNet, VggNet, LeNet, and sequential model on a dataset of 1500 images, achieving a 97% accuracy in predicting EB and LB.

In 2021, Islam Tarik et al. [58] used a CNN-based sequential model on a dataset of over 2034 images to diagnose potato diseases, achieving a 99.23% accuracy. In 2022, Bangal et al. [36] and Islam and Sikder [37] predicted EB, LB, and healthy potato classes using a CNN model with accuracies of 91.41% and 100% respectively, on the Plant Village dataset. In the same year, Kothari et al. [38] used DL to predict common potato diseases using 1200 images from the Plant Village dataset. Among various models, ResNet50 performed the best, achieving an accuracy of 98%. In 2021, Sinshaw et al. [67] used DL methods to identify LB in potato crops with imagery data. The InceptionV3 model achieved the highest accuracy at 87%, and other models like VGG16 and VGG19 were also tested. Biswas et al. 2014 [68] used DL-based image processing to predict LB occurrence. With 27 images and 340 regions of interest (ROIs), the Back Propagation Neural Network (BPNN) model was implemented and achieved a 93% accuracy rate. In 2021, Singh and Kaur [39] predicted three classes of potato diseases namely, EB, LB,

and healthy plants. An overall accuracy of 95% was achieved using the SVM model on a dataset of 54,306 images from the Plant Village dataset. In 2020, Iqbal and Talukder [40] used multiple ML approaches to predict EB and LB using 450 images from the Plant Village dataset. The research further tested several algorithms, including RF, LR, KNN, DT, NB, LDA (Linear Discriminative Analysis), and SVM. The RF model had the best performance with 97% accuracy. In 2021, Tiwari et al. [41] used CNN models like VGG16, VGG19, and InceptionV3 along with classifiers such as KNN, SVM, and LR to predict EB and LB. The LR model with VGG19 achieved an impressive 97.8% accuracy using a dataset of about 2,152 images. Dasgupta et al. 2020 [42] used DL methods to identify EB and LB, as well as healthy potato plants by applying the VGGNet model with transfer learning on 856 images from the Plant Village dataset, achieving a remarkable 100% accuracy. In 2016, Singh Brar and Singh [59] developed a method to classify potato crops as healthy or unhealthy using image processing and ML. A dataset of 100 potato images from a local market in India was collected. Fuzzy c-means clustering was operated and achieved an accuracy of 95%. In 2011, Razmjooy and Daviran [60] focused on predicting four different conditions of potato crops: healthy, unhealthy, standard, and malformed. The authors used DL-based image processing with models like SVM, RBF, and MLP neural networks. The SVM model performed exceptionally well with an accuracy of 99%. Rashid et al. [43] used a dataset of 4,062 images to predict healthy potatoes, EB, and LB. A CNN model was adopted and achieved a very high accuracy of 99.75%. In 2017, Islam et al. [44] used 300 images from the Plant Village dataset to predict EB, LB, and healthy potatoes. The SVM model demonstrated its effectiveness with an accuracy of 95%. Johnson et al. 2021 [69] used an imagery dataset of about 1,840 images to predict LB in potatoes. The study employed Mask R-CNN with a transfer learning approach, achieving an impressive accuracy of 98%. In 2020, Lee et al. [47] used DL-based image processing to differentiate between EB and LB diseases. A CNN model was employed for the Plant Village dataset and achieved an outstanding accuracy of 99%.

2) WEATHER-BASED POTATO DISEASE DETECTION

Kumar et al. 2024 [45] reported that the prevailing climate influenced potato leaf disease detection when weather data from 2006 to 2016 composed of 96453 instances along with plant disease data from Kaggle were used for the classification of EB, LB, and healthy classes. Different classifiers and regression models were used, and the RF classifier outperformed with 97% accuracy. Ibrahim et al. 2024 [61] explored meteorological and field data (2019-2021) to identify four classes of potato diseases namely, EB, LB, bacterial wilt, and healthy. The RF and MCC (Multi-Criteria Classification) were implemented for disease prediction and identifying disease severity. The accuracy of 67% – 97% was obtained based on the field points for different years. Sharma et al. [18] introduced an ML method to identify LB disease in potato

crops using weather data and information from the All India Coordinated Research Project (AICRP). The study implemented ANN, specifically a Multi-Layer Perceptron (MLP) to predict the disease with a notable accuracy of 90.9%. To boost the model's performance, the authors experimented with various activation functions like sigmoid, TanH, and ReLU. Fenu and Mallocci 2019 [70] used ML and DL to predict LB in potato crops in the Sardinia region, Italy. In this study, weather data from 2016 to 2018 (provided by ARPAS, the regional environmental agency) having 1074 instances was used. The researchers applied SVM and KNN classifiers and achieved high accuracy rates of 98% and 96%, respectively. In 2020, Fenu and Mallocci [71] implemented the SVM model to predict the risk of LB in the Sardinia region. The weather data from 2016 to 2019 was used to calculate a risk index and achieved an accuracy of 82%. Meno et al. 2022 [84] focused on predicting EB in potato crops using weather data collected over five years (2017-2021) from i-METOS in Limia, Spain. The researchers developed models based on plant growth and weather conditions, discovering that models using weather data were particularly effective. WE and GDD emerged as the most effective forecasting models based on plant-related factors for early symptom detection of EB. In 2023, Meno et al. [85] analyzed the relationship between weather conditions and sporangia (spore) levels in the air. The study used data from two growing seasons (2020 and 2021) to calculate Daily Risk Values for EB based on in-pressure, spore release, and weather conditions. Abderrahmane et al. 2022 [72] targeted LB detection in three regions: Ain Defla, Chlef, and Mostaganem in North-Western Algeria using weather data from 2012 to 2015. Auto Encoded Long Short-Term Memory Recurrent Neural Network (LSTM's AE) was deployed and an accuracy of 90.99% was achieved. Gu et al. 2016 [73] operated historical climate and disease occurrence data (1976-1985 and 2009-2012) to predict LB. A model called Blite_SVR was developed and achieved an accuracy of 64.3%. In 2015, Ghahreman et al. [86] used detailed weather data from 2013-2014 from i-Metos to predict EB in potato crops.

Henderson et al. 2007 [74] conducted in Southern Idaho made use of eight years of weather and disease severity data to predict LB with LR models and obtained an accuracy of over 70%. Singh et al. [75] utilized climate and crop data to distinguish LB in potatoes. The Extreme Learning Machine (ELM) model was used and an accuracy of 91.5% was observed. In 2017, the TOMCAST model proposed by Abuley et al. 2017 [87] was applied to anticipate EB in potatoes. The dataset was composed of climate and field data from 2015-2016. Escuredo et al. 2019 [88] made use of aerobiological samples and meteorological information to foresee EB in North-West Limia, Spain. Ahmed et al. 2015 [76] aimed to predict LB in potatoes in the Faisalabad area of Pakistan. The climate and disease severity data from 2011 to 2013 was utilized. The researchers used stepwise regression analysis and reported Root Mean Square Error (RMSE) (0.55), R^2 (0.80), and precision (6.0). Singh et al. [77] used

ML techniques to forecast LB using climate and AICRP data. The data from 1995 to 2018 was analyzed and models like ELM and SVM were operated. The ELM model achieved a maximum accuracy of 94.92%. In 2020, an innovative AI-based DL method, Fenu and Mallocci [78] introduced a decision support system to identify LB. The system used meteorological data for the disease risk index at three levels (low, medium, and high). Using the SimCast model for Blight Units and the FFNN ANN model, the study achieved 96% accuracy. In 2023, Choudhury et al. [62] used climate data to predict potato pests and diseases in West Bengal, India. The models like ANN, XGBoost, and RF were implemented. The study also evaluated the performance metrics like recall, precision, kappa coefficient, and F1-Score. Dar et al. [79] utilized meteorological data from 2017-2018 to forecast LB disease in potatoes using ML. The stepwise regression analysis was analyzed.

3) DISEASE DETECTION BASED ON VISUAL, MACHINE LEARNING, AND REMOTE SENSING

Fall et al. [80] explored a visual approach to detect LB in potato plants. The researchers observed 20 plants in a greenhouse to study the disease visually, providing a unique perspective on disease detection. In 2018, Duarte-Carvajalino et al. [81] combined ML, RS, and DL to forecast the disease severity index for potato LB. The data from 12 different potato varieties was employed for models such as MLP, CNN, SVR (Support Vector Regression), and RF. The study indicated that the CNN model outperformed the others. Hou et al. 2022 [82] focused on using ML to analyze images of potato plant leaves to detect LB. Distinct models, including SVM, KNN, and DT were employed and reported a high accuracy of 99%.

Recent advances in agricultural technology, especially ML, DL, and AI, have significantly improved how we manage crops. Hussain et al. 2020 [112] developed a smart sprayer using DL models like YOLOv3 to detect and selectively spray weeds and diseased potato plants. The developed sprayer reduced chemical use by 42-43%, lowering costs and environmental impact. Similarly, Farooque et al. 2023 [113] experimented with a smart sprayer that uses DL to find and treat weeds and diseased plants in potato fields and sprayer accurately targets plants needing treatment, cutting chemical use by about half while effectively weeds and diseases. Thus, advanced technologies make farming more efficient and environmentally friendly. ML has been applied in various fields beyond crop disease diagnosis, including cryptocurrency price prediction [114], where it is used to enhance predictive accuracy by integrating macroeconomic, microeconomic, and technical indicators.

D. MODEL PERFORMANCE MATRICES

The current section indicates that a variety of scientific methods have been adopted worldwide for disease forecasting. Most of this research has concentrated on classification tasks, with a significant emphasis on binary classification [28] and

multi-class classification [17]. Binary classification is typically used to determine the presence or absence of a disease using specific labels [28], [41], [50], [54]. In contrast, multi-class classification models [46], [47], [48], [51], [56] are employed to categorize diseases or predict the Disease Severity Level, with classifications ranging from 2 to 5 classes.

TABLE 1 provides detailed information on the 72 reviewed publications (research publications, journals, and conference publications), including references and years, datasets used, diseases identified, models employed, and metrics reported. This table serves as a comprehensive resource for other researchers by offering a structured overview of existing studies and facilitating easier access to relevant data. By summarizing key aspects of the literature, it aids in identifying research trends, gaps, and potential areas for future study.

To measure the accuracy of the proposed crop disease model, we need evaluation parameters that show the proposed model's performance. The various performance evaluation metrics found in the literature review are listed in TABLE 2.

This review includes an analysis of 72 publications (research/conference) and presents that advanced methods such as ML and DL are very beneficial for potato disease detection and identification. Among 72 publications, 51 publications presented potato disease detection using image processing and image classification techniques, while 18 publications utilized weather and disease data. Additionally, three publications sourced data from field collections and other sources. Thus, 70.8% of the existing research focused on image processing, whereas only 25% utilized climate data and 4.2% used other data sources. This disparity highlights a significant opportunity for researchers to future work in potato disease analysis using climate data.

The algorithms reviewed in these studies demonstrated a range of accuracies and performance based on the methodology and data used. For instance, using CNN, accuracy rates from 96.8% to 100% have been observed for potato leaf disease classification in studies [16], [17], [18], [19], [20], [27], [29], [32], [36], [38], [41], [42], [43], [48], [49], [50], [51], [52], [53], [77], [78], [80], [95], the most widely used CNN model called MobileNet utilized by researchers [23], [26], [46], [47] provided a maximum accuracy around 98.86% to minimum accuracy of 98.7%, again on imagery data.

VGG models including VGG16 and VGG19 been implemented [15], [23], [24], [33], [36], [37], [41], [42], [47], [50], [62], [80] on potato leaf images dataset and an accuracy range of 97% to 98.7% have been observed. Studies [23], [26], [36], [62] adopted a DL model called Inception and an average of 98.78% accuracy has been achieved. Several DNNs as mentioned in [23], [26], [52], [58] utilized imagery data for potato disease classification and achieved positive results. ResNet [15], [22], [33], [41], [42], [50] and EfficientNet [52] also demonstrated accuracy rates above 85% on imagery data.

Machine learning algorithms including SVM when applied to imagery data in [21], [30], [34], [36], [39], [44], [45], [49], [52], [55], [58], [67], and [78] and weather data in [66], [67], and [73], provides an accuracy rate from 82% to 99.75%.

TABLE 1. Summary of reviewed publications: References, Datasets, Diseases Identified, Models, and Metrics (2007-2024).

S. No	References & Year	Dataset	Disease	Model	Metrics
1.	[17], 2024	Imagery	Early Blight, Late Blight, & Healthy	CNN	Accuracy: 97.4%
2.	[27], 2024	Imagery	Early Blight, Late Blight, & Healthy	MDSCIRNet with SVM, LR, AdaBoost, and RF	Accuracy: 99.33%
3.	[29], 2024	Imagery	Early Blight, Late Blight, & Healthy	MobileNetV2, VGG16, VGG19, InceptionV3, and Xception	Accuracy: 98.7%
4.	[61], 2024	Meteorological and Field Data	Early Blight, Late Blight, Bacterial Wilt, & Healthy	RF and MCC	Accuracy: 62 – 96%
5.	[52], 2024	Imagery	Viruses, Bacteria, Fungi, Pests, Nematodes, Phytophthora, and Healthy	EfficientNetV2B3, MobileNetV3-Large, and VGG16 (With or Without Data Augmentation)	Accuracy: 73.63%
6.	[31], 2024	Imagery	Early Blight, Late Blight, & Healthy	RF, NB, and KNN	Accuracy: 99%
7.	[24], 2024	Imagery	Early Blight, Late Blight, & Healthy	CNN	Accuracy: 99.59%
8.	[53], 2024	Imagery	Common Scab, Black Scurf, Blackleg, Dry Rot, Pink Rot, Miscellaneous, and Healthy	CNN custom model ConvNeXtSmall Model	Accuracy: 99.49%
9.	[32], 2024	Imagery	Early Blight, Late Blight, & Healthy	DNN Ensemble Models which integrates ResidualNet, MobileNet, and Inception Models	Accuracy: 98.86%
10.	[26], 2024	Imagery	Early Blight, Late Blight, & Healthy	CNN	Accuracy: 97.33%
11.	[25], 2024	Imagery	Early Blight, Late Blight, & Healthy	CNN	Accuracy: 97.82%
12.	[45], 2024	Weather Data	Early Blight, Late Blight, & Healthy	RF Classifiers	Accuracy: 97%
13.	[51], 2023	Imagery	15 Classes for Tomato, Potato, And Bell Pepper	DenseNet201	Accuracy: 98.67%
14.	[23], 2023	Imagery	Early Blight, Late Blight, & Healthy	CNN	Accuracy: 96.8%
15.	[28], 2023	Imagery	Early Blight, Late Blight, & Healthy	ResNet-9	Accuracy: 99.25%
16.	[30], 2023	Imagery	Early Blight, Late Blight, & Healthy	ODCNN and VGG19	Accuracy: 99.22%
17.	[55], 2023	Imagery	Healthy and Afflicted Specimens	CNN Models: VGG19, DenseNet121, and ResNet50	Accuracy: 98.77%

TABLE 1. (Continued.) Summary of reviewed publications: References, Datasets, Diseases Identified, Models, and Metrics (2007-2024).

18.	[112], 2020	Imagery	Weeds, Early Blight, and Healthy	YOLOv3	-
19.	[85], 2023	9 Cultivators of Potato (2021-2022)	Early Blight	IP, DRV and SR	Sporangia Concentration for T:10C and RH: 80%
20.	[56], 2023	Imagery	Healthy, Black Scurf, Common Scab, Black Leg, and Pink Rot	CNN	Accuracy: 100%
21.	[50], 2023	Imagery	Early Blight and Late Blight	Lightweight NN	Accuracy: 93%
22.	[57], 2023	Imagery	-	M-SVM and CNN	Accuracy: 99%
23.	[63], 2023	Imagery	-	DNN and M-SVM + CNN	Accuracy: 99%
24.	[33], 2023	Imagery	Late Blight, Early Blight, and Healthy	Sequential Model (Based on CNN)	Accuracy: 94.2%
25.	[34], 2023	Imagery	Early Blight and Late Blight	CNN	Accuracy: 98% and Loss: 0.04
26.	[15], 2023	Imagery	Early Blight, Late Blight, and Healthy	CNN	Accuracy: 97%
27.	[62], 2023	Climate and Imagery Data	Thrips, Early Blight, Late Blight, Septoria Leaf Spot, and Healthy	Baseline Models and Enhanced Baseline Models (ANN, XNG, and RF)	Precision, Recall, F1-Score and Kappa Coefficient
28.	[22], 2022	Imagery	Early Blight, Late Blight, & Healthy	K-Means Segmentation, VGG16, VGG19, and ResNet50	Accuracy: 97%
29.	[84], 2022	Weather Data from iMETOS	Early Blight	Early Blight forecast models	-
30.	[72], 2022	Climate and LB Data	Late Blight	LSTM-AEs Models	Accuracy: 90.99%
31.	[82], 2022	Imagery	Late Blight	SVM, KNN, and DT	Accuracy: 99%
32.	[64], 2022	UAV- based hyperspectral imagery (3 experimental sites)	Late Blight	CropDocNet DL	Accuracy: 98.08%
33.	[65], 2022	Imagery	Late Blight	K-Means and Hierarchical clustering (3 Clusters)	Accuracy: 68.24%
34.	[35], 2022	Imagery	Early Blight, Late Blight, and Healthy	SVM, KNN, LR, and ANN	Accuracy: 99.75%
35.	[36], 2022	Imagery	Late Blight, Early Blight, and Healthy	CNN	Accuracy: 91.41%
36.	[37], 2022	Imagery	Early Blight, Late Blight, and Healthy	CNN	Accuracy: 100%
37.	[38], 2022	Imagery	Early Blight, Late Blight, and Healthy	GoogleNet, ResNet50 and VGG16	Accuracy: 97%
38.	[16], 2021	Imagery	Early Blight	CNN Models: GoogleNet, VGGNet, & EfficientNet	F1Score: 0.84 - 0.98

TABLE 1. (Continued.) Summary of reviewed publications: References, Datasets, Diseases Identified, Models, and Metrics (2007-2024).

39.	[48], 2021	Imagery	Early Blight, Late Blight, & Healthy	K-Means Clustering, SVM, KNN, NB, and DT	Accuracy: 92.9%
40.	[54], 2021	Imagery	-	K-Means Clustering, CNN, and SVM	Accuracy: 97.90%
41.	[58], 2021	Imagery	Potato Leaf Roll Virus, Hollow Heart, Scab, Early Blight, Soft Rot, Sotali Poka Rog, and Virus Jonito Rog	CNN-based Sequential Model	Accuracy: 99.23%
42.	[67], 2021	Imagery	Late Blight	InceptionV3, VGG16 and VGG19	Accuracy: 87%
43.	[39], 2021	Imagery	Late Blight, Early Blight, and Healthy	M-SVM	Accuracy: 95.99%
44.	[79], 2021	Weather Data (2017-2018)	Late Blight	Stepwise Regression	R ² and RMSE
45.	[43], 2021	Imagery	Early Blight, Late Blight, and Healthy	CNN	Accuracy: 99.7%
46.	[69], 2021	Imagery	Late Blight and Healthy	Mask R-CNN using Transfer Learning	Accuracy: 98%
47.	[71], 2020	Weather Data (2016-2019)	Late Blight	SVM	Accuracy: 82%
48.	[46], 2020	Imagery	Early Blight, Late Blight, and Healthy	CNN Models: AlexNet, ResNet, VggNet, LeNet, and Sequential Model	Accuracy: 97%
49.	[78], 2020	Weather Data	Late Blight	FFNN	Accuracy: 96%
50.	[40], 2020	Imagery	Early Blight, Late Blight, and Healthy	RF, LR, KNN, DT, NB, LDA, and SVM	Accuracy: 97%
51.	[41], 2020	Imagery	Early Blight, Late Blight, and Healthy	Models: VGG16, VGG19 and InceptionV3 Classifiers: KNN, SVM, and LR	Accuracy: 97.8%
52.	[42], 2020	Imagery	Early Blight, Late Blight, and Healthy	VGGNet with Transfer Learning	Accuracy: 100%
53.	[47], 2020	Imagery	Early Blight, Late Blight, and Healthy	CNN	Accuracy: 99%
54.	[70], 2019	Weather Data	Late Blight	ANN and SVM	Accuracy: 98%
55.	[88], 2019	8 Years Aerobiological Samples and Meteorological Data	Early Blight	Spearman Correlation Analysis	-
56.	[77], 2019	Climate and AICRP Data	Late Blight	SVM and ELM	Accuracy: 94.92%
57.	[18], 2018	Weather Data	Late Blight	ANN	Accuracy: 90.9%
58.	[81], 2018	Agriculture Field Dataset for a Region of 12 Varieties	Late Blight	MLP, CNN, SVM, and RF	RMSE: 15.09, R2: 0.75 and MAE: 12.96

TABLE 1. (Continued.) Summary of reviewed publications: References, Datasets, Diseases Identified, Models, and Metrics (2007-2024).

59.	[75], 2018	Weather and AICRP Potato Database	Late Blight	ELM	Accuracy: 91.5%
60.	[87], 2017	Weather and Field Data for Disease Assessment	Early Blight	TOMCAST Models	-
61.	[49], 2017	Imagery	Late Blight and Early Blight	KNN, RF, and SVM	Accuracy: 92%
62.	[44], 2017	Imagery	Late Blight, Early Blight, and Healthy	SVM	Accuracy: 95%
63.	[73], 2016	Climate and Disease 1 st date occurrence	Late Blight	BLITE – SVR	Accuracy: 64.3%
64.	[59], 2016	Imagery	Healthy and Unhealthy	Fuzzy C-Mean Clustering	Accuracy: 95%
65.	[80], 2015	20 Plants in Greenhouse	Late Blight	Visual Basis	-
66.	[86], 2015	Hourly Meteorological data (from iMetos)	Early Blight	PD (Phenological Day) and DD (Degree Day)	Differentiate Between the Actual Dates and Observed Dates
67.	[76], 2015	Climate and Potato LB Severity Data	Late Blight	Stepwise Regression Analysis	MSE: 0.55, R ² : 0.80 and P: 6.0
68.	[83], 2014	Satellite Imagery	Late Blight	Remote Sensing Indices	-
69.	[68], 2014	Imagery	Late Blight	BPNN	Accuracy: 93%
70.	[66], 2011	Hyperspectral Data	Late Blight	Vegetational Indices: NDVI, SR, and SAVI	Reflectance Curved
71.	[60], 2011	Imagery	Healthy, Unhealthy, Standard, and Malformed Potatoes	3 Neural Networks: SVM, RBF, and MLP	Accuracy: 99 %
72.	[74], 2007	5 Regions Weather Data with disease severity	Late Blight	Binary Logistic Regression and Logistic Linear Regression	Accuracy: 70%

Another most widely used ML algorithm, RF [21], [40], [56], [77] when applied on diverse datasets offers an average accuracy of 85.67%. KNN algorithm utilized in [25], [30], and [36] provides an accuracy of 99%. The maximum observed accuracy across the studies was 100%, while the minimum was 64.3%, and an overall average accuracy of 94.7%. These results emphasize the potential of ML and DL models in achieving high accuracy in potato disease classification. The diversity in the performance of different models also indicates that the choice of model should be modified to the nature of the dataset and the type of disease being detected.

FIGURE 9 provides a visual summary of the distribution of various ML and DL techniques used in the reviewed

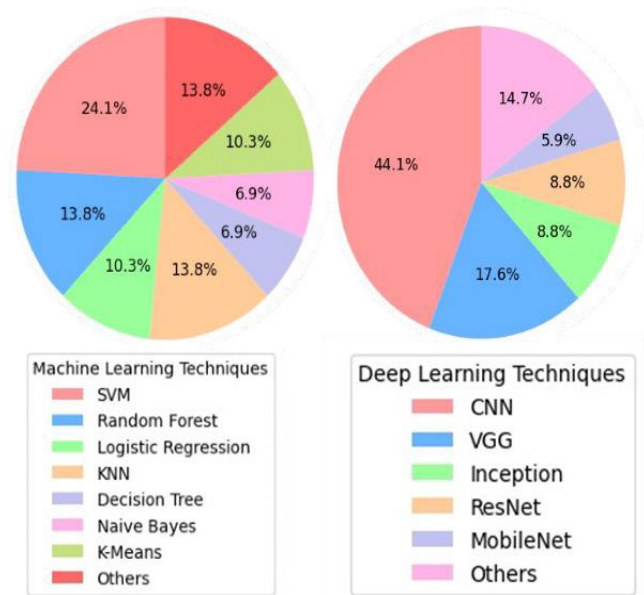
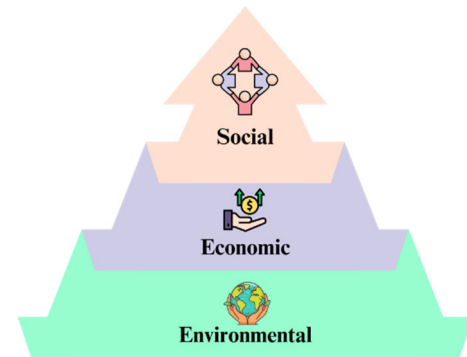
studies. The first pie chart represents the distribution of ML techniques, with SVM (24.1%) being the most frequently used, followed by RF and KNN (13.8%), LR and K-Means (10.3%), DT and NB (6.9%), and others (13.8%) algorithms. The second pie chart shows the distribution of DL techniques, with CNN (44.1%) being the predominant method, followed by VGG (17.6%), Inception and ResNet (8.8%), MobileNet (5.9%), and others (14.7%).

In FIGURE 10, the pyramid emphasizes the benefits of implementing advanced technologies in agriculture, which align with social, economic, and environmental sustainability goals. Socially, accurate disease detection helps ensure food security and supports farmer's livelihoods. Economically, it reduces crop losses and minimizes the

TABLE 2. Performance metrics reference table.

S. No.	Metrics	Formulas
1.	Accuracy (Acc)	$\frac{TP + TN}{TP + TN + FP + FN}$
2.	Precision (Prec)	$\frac{TP}{TP + FP}$
3.	Recall (Rec)	$\frac{TP}{TP + FN}$
4.	Specificity	$\frac{TN}{TN + FP}$
5.	F1- Score	$\frac{2TP}{2TP + FP + FN}$
6.	Mean Square Error (MSE)	$\frac{1}{N} \sum Actual - Predicted ^2$
7.	Root Mean Square Error (RMSE)	$\frac{\sqrt{\sum_{i=1}^N (Predicted_i - Actual_i)^2}}{N}$
8.	R Square	$1 - \frac{SSR}{SST}$
9.	Mean Average Precision (MAP)	$\left(\frac{1}{No. of Classes}\right) \sum_c Precision_c$
10.	Receiver Operating Characteristic Curve (ROC)	$\frac{\sqrt{(Sensitivity^2 + Specificity^2)}}{\sqrt{2}}$
11.	Mean Absolute Error (MAE)	$\frac{1}{N} \sum Actual - Predicted $
12.	Mean Average Percentage Error (MAPE)	$\frac{100\%}{N} \sum \left \frac{Actual - Predicted}{Actual} \right $
13.	Mean Percentage Error (MPE)	$\frac{100\%}{N} \sum \left(\frac{Actual - Predicted}{Actual} \right)$
14.	Error Rate	$\frac{(FP + FN)}{(P + N)}$
15.	Mean Intersection of Union (MIoU)	$\left(\frac{1}{No. of Classes}\right) * \sum_c IoU_c$

cost associated with chemical treatments. Environmentally, it promotes sustainable farming practices by reducing the

**FIGURE 9.** Distribution of most widely used machine learning (left) and deep learning techniques (right) in reviewed publications.**FIGURE 10.** Key pillars of sustainability: social, economic, and environmental.

overuse of pesticides and enhancing crop health through early intervention.

Thus, the integration of machine learning and deep learning models in potato disease detection holds significant promise. Future research should explore the potential of using climate data to complement image processing techniques, address the current gap in this area, and leverage the strengths of both data types to enhance disease management in agriculture. Additionally, advanced approaches in ML and DL for potato disease forecasting should be pursued. Geographical dissimilarities can impart variations in model accuracy, indicating the need for more extensive data to create more universal crop disease prediction models. More research is required to develop disease forecasting software to provide first-hand information to farmers for safeguarding the crops. The advanced disease prediction approaches can provide valuable insights into disease trends and assist farmers and growers in timely interventions, ultimately contributing to improved agricultural productivity and sustainability. In addition to the points discussed in the manuscript, another

important aspect to consider is the limitations and ethical considerations of AI in agriculture, particularly regarding accessibility and scalability. For instance, in regions like North America, where farm sizes are larger, AI-based technologies can be more easily adopted and integrated into sustainable farming practices. However, in countries like India and Pakistan, where farm sizes are smaller, the implementation of AI models is often less feasible, and traditional farming practices remain more prevalent. Scalability is a key challenge in these areas, as AI solutions may not effectively cater to small-scale farming operations. Thus, the paper also emphasizes the importance of not overshadowing traditional farming practices when integrating AI solutions. Furthermore, farmers are essential stakeholders in the development of AI models, and their input, alongside that of industry partners and experts, is vital in creating ethical and practical AI applications. While AI offers advancements in productivity, disease detection, and resource management, it is essential to balance these innovations with respect for traditional methods that may still hold significant value, particularly in unique environmental or socio-economic contexts. AI technologies should complement traditional farming practices rather than replace them entirely. It's crucial that these solutions are tailored to the local context and respect the valuable knowledge farmers have built over time. This balanced approach would help bridge the gap between modern and traditional techniques, allowing farmers to adopt AI tools in a way that enhances, rather than disrupts, their existing farming systems. Ensuring this balance is critical for the ethical deployment of AI, as it promotes inclusivity and respects the diverse practices that have sustained agriculture in various regions for centuries.

IV. CONCLUSION AND RECOMMENDATIONS

The problem statement addressed in this review is the critical need for accurate and efficient potato disease detection and forecasting to ensure agricultural sustainability and food security. The review highlights significant improvements in this field, demonstrating that ML and DL methods have already achieved high accuracy rates in potato disease classification. For instance, CNN has shown promising accuracy rates between 96.8% to 100% for detecting the most common potato diseases like EB and LB. Other models, such as SVM, RF, KNN, DT, KNN, MobileNet, VGG, Inception, and ResNet, have also demonstrated notable performance. Potato disease forecasting is of utmost importance due to the substantial impact of potato crops on global food security. Accurate disease detection helps farmers make timely interventions, reducing crop losses and minimizing the use of chemical treatments. This optimal use of resources not only benefits farmers economically but also promotes sustainable farming practices by reducing pesticide overuse and enhancing crop health through early prevention.

Despite the advancements, the field is still in its infancy, and more research is needed. The review identifies a significant gap in the use of climate data for disease analysis,

with only 25% of the existing research utilizing weather data. Future research should be more focused on exploring the potential climate to enhance disease management in agriculture. Moreover, time-series forecasting methods should be looked at to improve disease analysis. This future research addresses gaps present in existing research like data unavailability, geographical variations, and scalability issues. The ML and DL models, implemented in the literature point out the need to opt for the right model based on the dataset and type of disease to be identified. The replacement of conventional methods by advanced methods offers valuable insights into disease patterns and assists farmers in making informed decisions. These advancements support agricultural sustainability, food security, and human well-being.

In conclusion, ML and DL models for potato disease detection hold meaningful importance. More research is required to develop robust disease forecasting models and applications to ensure timely information to the farmers and growers and help with quick actions. This review emphasizes the need for ongoing research and new ideas in this field, which will lead to enhanced potato productivity and food security.

ACKNOWLEDGMENT

The authors would like to thank the Natural Sciences and Engineering Council of Canada for supporting the project.

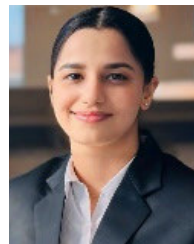
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